

A. Further Details on the Experimental Setup

A.1. Model Specification

The following table lists the models and their corresponding links.

Models	Links
Llama-2-7B (Touvron et al., 2023b)	https://huggingface.co/meta-llama/Llama-2-7b-hf
Llama-2-7B-Chat (Touvron et al., 2023b)	https://huggingface.co/meta-llama/Llama-2-7b-chat-hf
Llama-2-13B-Chat (Touvron et al., 2023b)	https://huggingface.co/meta-llama/Llama-2-13b-chat-hf
Llama-2-70B-Chat (Touvron et al., 2023b)	https://huggingface.co/meta-llama/Llama-2-70b-chat-hf
Llama-3-8B (AI@Meta, 2024)	https://huggingface.co/meta-llama/Meta-Llama-3-8B
Llama-3-8B-Instruct (AI@Meta, 2024)	https://huggingface.co/meta-llama/Llama-2-7b-hf
Llama-3-70B-Instruct (AI@Meta, 2024)	https://huggingface.co/meta-llama/Meta-Llama-3-70B-Instruct
Mistral-7B (Jiang et al., 2023)	https://huggingface.co/mistralai/Mistral-7B-v0.1
Mistral-7B-Instruct (Jiang et al., 2023)	https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.2
Qwen2-7B (Team, 2024a)	https://huggingface.co/Qwen/Qwen2-7B
Qwen2-7B-Instruct (Team, 2024a)	https://huggingface.co/Qwen/Qwen2-7B-Instruct
Qwen2-72B-Instruct (Team, 2024a)	https://huggingface.co/Qwen/Qwen2-72B-Instruct
Qwen2.5-1.5B-Instruct (Team, 2024b)	https://huggingface.co/Qwen/Qwen2.5-1.5B-Instruct
Qwen2.5-32B-Instruct (Team, 2024b)	https://huggingface.co/Qwen/Qwen2.5-32B-Instruct
Qwen2.5-72B-Instruct (Team, 2024b)	https://huggingface.co/Qwen/Qwen2.5-72B-Instruct

A.2. Dataset Specification

The following table lists the datasets and their corresponding links.

Datasets	Links
Ultrachat (Ding et al., 2023)	https://huggingface.co/datasets/stingning/ultrachat
AlpacaEval (Taori et al., 2023)	https://huggingface.co/datasets/tatsu-lab/alpaca
GSM8K (Cobbe et al., 2021)	https://huggingface.co/datasets/openai/gsm8k
MATH (Saxton et al., 2019)	https://huggingface.co/datasets/deepmind/math_dataset
TinyStories (Eldan & Li, 2023)	https://huggingface.co/datasets/roneneldan/TinyStories
ROCStories (Mostafazadeh et al., 2016)	https://huggingface.co/datasets/Ximing/ROCStories

CommonsenseQA (Talmor et al., 2019)	https://huggingface.co/datasets/tau/commonsense_qa
SocialIQA (Sap et al., 2019)	https://huggingface.co/datasets/allenai/social_i_qa
MedMCQA (Pal et al., 2022)	https://huggingface.co/datasets/openlifescienceai/medmcqa
ARC-Challenge (Clark et al., 2018)	https://huggingface.co/datasets/allenai/ai2_arc
CREAK (Onoe et al., 2021)	https://huggingface.co/datasets/amydeng2000/CREAK
FEVER (Thorne et al., 2018)	https://huggingface.co/datasets/fever/fever

A.3. Detailed Process of Response Collection and Labeling

In this section, we detail the process of collecting a dataset $\mathcal{D} = \{\mathcal{H}_i, \hat{g}_i\}_{i=1}^N$ for each task $T = (p(\mathbf{x}), g(\mathbf{y}))$, pairing prompt representations with their corresponding attribute labels. First, we construct the prompt distribution $p(\mathbf{x})$ to elicit responses with target attributes from the models (Sec.A.3.1). Second, we label these responses according to specific criteria $\hat{g}_i = g(\mathbf{y}_i)$ to capture their key attributes (Sec.A.3.2). Finally, we collect representations $\mathcal{H}_i = \{\mathbf{H}_{\mathbf{x}_i}^l\}_{l=1}^L$ for each prompt (Sec. A.3.3).

A.3.1. PROMPT TEMPLATES

To elicit responses with target attributes, we construct prompt distributions using carefully designed templates paired with datasets. We present the prompt templates for both chat and base models across all tasks, along with representative input-output examples.

Task 1: Response Length
Prompt for fine-tuned models
''
{data}
''
(→ Gets formatted according to model’s template)
Example Response
Data: Why are oceans important to the global ecosystem?
Output: The oceans play a crucial role [...]
Prompt for base models
''
Q: How can cross training benefit athletes?
A: Cross training offers various benefits [...] [END OF RESPONSE]
Q: What role does collaboration play in creativity?
A: Collaboration and originality complement each other [...] [END OF RESPONSE]
Q: {data}
A:
''
Example Response
Data: What are positive impacts of Reality TV?
Output: Reality TV provides entertainment and [...] [END OF RESPONSE]
Task 2: Reasoning Steps
Prompt for fine-tuned models
''
Provide step-by-step solution, starting with 'Step 1:'.
Problem:
{data}

''

(→ Gets formatted according to model's template)

Example Response

Data: Randy has 60 mango trees on his farm. He also has 5 less than half as many coconut trees as mango trees. How many trees does Randy have in all?

Output: Step 1: Write down the information [...]

Prompt for base models

''

Solve this problem step-by-step, starting with 'Step 1:'.

Few-shot examples:

Problem: Let $f(x) = \begin{cases} ax+3 & \text{if } x > 2 \\ x-5 & \text{if } -2 \leq x \leq 2 \\ 2x-b & \text{if } x < -2 \end{cases}$.

Find $a+b$ if f is continuous.

Step 1: At $x=2$: $a(2)+3=2-5$ [...] [END OF RESPONSE]

Problem: If $x=2$ and $y=5$, find $(x^4+2y^2)/6$.

Step 1: Substitute: $(2^4+2(5^2))/6$ [...] [END OF RESPONSE]

Problem: {data}

''

Example Response

Data: Weng earns \$12 an hour for babysitting. Yesterday, she just did 50 minutes of babysitting. How much did she earn?

Output: Step 1: Substitute: $12(50/60)$ [...] [END OF RESPONSE]

Task 3: Character Choices

Prompt for fine-tuned models

''

Here's the first sentence of a story: {data}

Continue this story with one sentence that introduces a new animal character.

''

(→ Gets formatted according to model's template)

Example Response

Data: Once upon a time, there was a big car named Dependable.

Output: As Dependable was cruising down the highway, a chatty parrot [...]

Prompt for base models

''

First sentence: Lily was a little mouse who liked to follow her big brother Leo.

Continuation: The garden was peaceful that morning until [...]

[Animal: owl] [END OF RESPONSE]

First sentence: Lila and Ben were playing in the park with their toys.

Continuation: While building their epic sandcastle [...] [Animal: rabbit] [END OF RESPONSE]

First sentence: Sara was lonely.

Continuation: As she sat on the front steps drawing patterns [...]

[Animal: puppy] [END OF RESPONSE]

First sentence: Lily and Ben were twins who liked to go on walks with their mom and dad.

Continuation: Their morning hike through the woods [...] [Animal: squirrel] [END OF RESPONSE]

First sentence: {data}

Continuation: